

# Route decision: Shanghai Global Distribution Center → Rotterdam Distripark

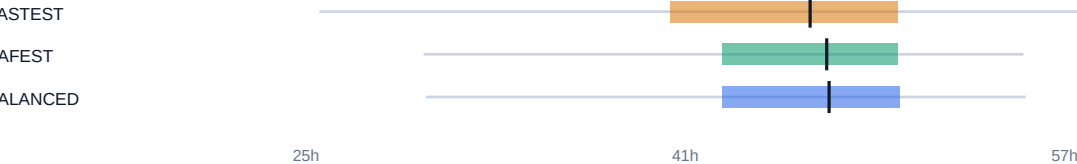
HUB-SHANGHAI → HUB-ROTTERDAM · mode any (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC · cargo value ¥1,573,151,090 at 25%/yr carrying cost

**Recommendation: BALANCED**, AIR via direct. Expected arrival in **42 h** (1.7 days); plan for 46 h (p85), worst case 49 h (p95). Cost: freight ¥21,030,318, landed ¥22,911,454 incl. ¥1,881,136 carrying cost. Peak leg risk 2%.

## Options

| Option   | Mode | Via    | Plan h | p50 h | p85 h | p95 h | Freight     | Landed      | Risk |
|----------|------|--------|--------|-------|-------|-------|-------------|-------------|------|
| FASTEST  | AIR  | direct | 25     | 40    | 46    | 49    | ¥21,830,585 | ¥23,612,951 | 2%   |
| SAFEST   | AIR  | direct | 29     | 42    | 46    | 49    | ¥21,927,470 | ¥23,808,607 | 2%   |
| BALANCED | AIR  | direct | 29     | 42    | 46    | 49    | ¥21,030,318 | ¥22,911,454 | 2%   |

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



## Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 39.7 h (p85 45.6 h, p95 49.3 h), ¥23,612,951 landed, 2 handoffs. 2.2 h faster than the next option. ¥701,468 more (landed) than BALANCED.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost        | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|-------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | ¥5,659      | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 3 | AIR     | Frankfurt Airport             | 8,858 | 11.1  | +8.8      | ¥20,902,476 | 0%   |        |
| 4 | AIR     | Amsterdam Airport Cargo       | 366   | 0.5   | +8.8      | ¥863,169    | 0%   |        |
| 5 | handoff | Amsterdam Airport Cargo       |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 64    | 0.9   | +3.6      | ¥12,087     | 0%   |        |

Transit 12.9 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 24.9 h. Model delay buffer p50/p85/p95: +14.7 / +24.9 / +32.0 h. Route band p50 39.7 h, p85 45.6 h, p95 49.3 h (independent legs); 56.9 h if all legs are late together. Costs: linehaul ¥21,783,390, transfers ¥47,195, disruption surcharge ¥0. Inventory carrying cost ¥1,782,366 expected (¥2,213,365 at p95); landed ¥23,612,951.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.3 h, p95 49.3 h), ¥23,808,607 landed, 2 handoffs. 2.2 h slower than FASTEST. ¥897,168 more (landed) than BALANCED.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost        | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|-------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | ¥5,659      | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 3 | AIR     | London Heathrow Airport       | 9,241 | 11.6  | +8.8      | ¥21,806,622 | 0%   |        |
| 4 | handoff | London Heathrow Airport       |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 5 | ROAD    | Calais Channel Tunnel Hub     | 169   | 2.5   | +3.6      | ¥31,929     | 0%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 191   | 2.8   | +3.6      | ¥36,066     | 0%   |        |

Transit 17.3 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.3 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.3 h, p95 49.3 h (independent legs); 54.6 h if all legs are late together. Costs: linehaul ¥21,880,276, transfers ¥47,195, disruption surcharge ¥0. Inventory carrying cost ¥1,881,136 expected (¥2,213,365 at p95); landed ¥23,808,607.

Why the slowest leg (AIR-SHANGHAI → AIR-HEATHROW, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.4 h, p95 49.4 h), ¥22,911,454 landed, 2 handoffs. 2.2 h slower than FASTEST. ¥701,468 cheaper (landed) than the lowest-cost alternative.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost        | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|-------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | ¥5,659      | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 3 | AIR     | Frankfurt Airport             | 8,858 | 11.1  | +8.8      | ¥20,902,476 | 0%   |        |
| 4 | handoff | Frankfurt Airport             |       | 6.0   | +0.0      | ¥23,597     | 2%   |        |
| 5 | ROAD    | Duisburg Intermodal Terminal  | 200   | 2.9   | +3.6      | ¥37,828     | 0%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 197   | 2.9   | +3.6      | ¥37,161     | 0%   |        |

Transit 17.4 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.4 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.4 h, p95 49.4 h (independent legs); 54.7 h if all legs are late together. Costs: linehaul ¥20,983,123, transfers ¥47,195, disruption surcharge ¥0. Inventory carrying cost ¥1,881,136 expected (¥2,217,855 at p95); landed ¥22,911,454.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.